

Suggested FYP topics for 2026-27

by

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Project code :	ZYY1
Project title :	LLM Agent-empowered Robot Sensing System
No. of students :	1-2
Description of project :	Embodied Artificial Intelligence (AI) is an emerging field poised to revolutionize daily life by integrating advanced reasoning into physical machines. Recent breakthroughs in Large Language Models (LLMs) and agentic systems now provide robots with a powerful "brain" capable of deep contextual understanding and complex decision-making. However, for these LLM agents to reason effectively, they require a robust mechanism to perceive their physical surroundings. This Final Year Project aims to design and implement an LLM agent-empowered robot sensing system. The core objective is to develop a comprehensive data pipeline that captures real-world environmental data, processes it, and seamlessly translates it into a format the LLM can interpret for downstream reasoning and task execution. Students will set up the hardware by integrating sensors onto a robotic platform and build the software interface connecting these sensors to the LLM agent. Ideal candidates should possess a strong background in machine learning and/or robotics programming. This project offers a unique opportunity to work at the cutting edge of AI, bridging the gap between digital intelligence and physical perception.
Industrial collaboration :	No.

Project code :	ZYY2
Project title :	Smart 3D Indoor Navigation App for the CUHK Campus
No. of students :	1-2
Description of project :	Navigating the complex, multi-level terrain and intricate building layouts of the CUHK campus can be a daunting experience for newcomers, with hidden elevators and confusing walkways often difficult to locate. To solve this challenge, this Final Year Project aims to develop a comprehensive 3D indoor navigation smartphone application tailored specifically for the CUHK campus. The core objective is to create an intuitive mobile app that accurately estimates a user's indoor location and generates optimal, point-to-point navigation plans, seamlessly guiding them through building interiors and elevator networks. Students will be responsible for designing the 3D map interface, integrating indoor positioning algorithms, and developing the routing logic. Candidates are expected to have a strong interest in mobile software engineering,

	possessing or demonstrating a strong willingness to learn iOS or Android app development frameworks and UI design. This project offers excellent practical experience in mobile computing while delivering a highly impactful tool to improve campus accessibility.
Industrial collaboration :	No.

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